

PROJECT STATUS REPORT

Trading Backtester

Current Standing, Remaining Roadmap, Conceptual Knowledge Map and Safe Engineering Implementation Plan

Repository checkpoint	b37ef0c - Complete EURUSD data validation and resampling
Blueprint basis	Python Trading Strategy Backtester - Architecture and Sprint Plan
Assessment date	29 August 2026
Current verified tests	15 passing
Current project description	Tested historical market-data foundation; strategy and execution layers remain placeholders

Executive conclusion

The data foundation is substantially complete, but the strategy engine has not started. Three of the four committed Sprint 1 stories are complete. The project is approximately 75% through Sprint 1 and approximately 15-20% through the full six-sprint blueprint.

This report documents status and safe engineering work. It does not provide trade-entry, stop, target, realistic execution, or P&L; implementation instructions.

1. Current Standing

The repository review shows that the implemented work is concentrated in ingestion, cleaning, validation, resampling, testing, logging, and documentation. The model, indicator, strategy, execution, and backtest packages remain empty or structural placeholders.

Blueprint milestone	Status	Evidence
Project foundation	Complete	Environment, folders, configuration, logging, requirements, Git and documentation
Historical data pipeline	Complete	Twelve Data ingestion, UTC normalization, cleaning, diagnostics and Parquet storage
EUR/USD historical acquisition	Complete	2,456,032 unique one-minute rows to earliest accessible API boundary
15-minute resampling	Complete	157,553 complete UTC-aligned candles
Automated data tests	Complete	Nine cleaner tests and six resampler tests
Swing engine	Not started	Swing model and indicator module are empty
15-minute market structure	Not started	No active-swing or state-transition implementation
Lower-timeframe pattern layer	Not started	No imbalance-zone or chronological confirmation engine
Volatility and risk layer	Not started	ATR period exists in settings, but no calculator uses it
Simulation and analytics	Not started	Backtest and execution packages are placeholders

Completion interpretation

- Sprint 1: about 75% complete because Stories 1-3 pass their acceptance criteria and Story 4 is missing.
- Entire blueprint: about 15-20% complete because the foundational work is strong but Sprints 2-6 remain mostly unimplemented.
- Data engineering: about 85-90% complete for EUR/USD, with generalization and integrity hardening still needed.
- Strategy and simulation functionality: effectively 0% complete.

2. Verified Data Checkpoint

Metric	Verified value
Raw rows	2,456,032
Cleaned rows	2,454,877
Rounding inconsistencies repaired	1,432
Corrupt OHLC rows rejected	1,155
Timestamp gaps after cleaning	11,501
Complete 15-minute candles	157,553
One-minute range	2020-04-07 06:54 UTC to 2026-08-29 13:29 UTC
The cleaner repairs only EUR/USD boundary inconsistencies no larger than 0.00002 and rejects larger violations. Missing candles are preserved as missing; the system does not fabricate prices.	

3. Blueprint-by-Blueprint Gap Analysis

Sprint 1 - Foundation and Swings

Foundation, ingestion and resampling are complete. The missing committed story is a chronological pivot model with separate event time and confirmation time, tests proving delayed knowledge, and manual historical reconciliation. The optional market-structure prototype is also absent.

Sprint 2 - Higher-Timeframe Structure

The active-observation lifecycle, close-versus-wick classification, temporary pending state, next-period resolution, session-time conversion, pre-session availability, and transition logging are all unfinished.

Sprint 3 - Lower-Timeframe Pattern Research

The project lacks the lower-timeframe chronological framework, three-candle discontinuity classification, zone lifecycle, interaction classification, and event-order verification.

Sprint 4 - Volatility and Simulation

No rolling volatility calculator, instrument conversion layer, lifecycle simulator, ambiguity policy, or result schema is implemented. The repository setting `ATR_PERIOD=20` is currently unused and differs from the blueprint's unresolved-period note.

Sprint 5 - Advanced Structural Analysis

Untouched-level lifecycle, second qualifying structure, objective round-level significance, historical reaction criteria, and broader manual validation are all open.

Sprint 6 - Analytics

Event logs, descriptive performance reporting, database output, discrepancy reports, and visualization remain open.

4. Unresolved Specification Decisions

Decision	Why it blocks reliable implementation
Rolling volatility period	The blueprint leaves the period unconfirmed while the repository currently stores 20. A configured value is not the same as a validated rule.
Structural range ownership	Edge cases do not specify which confirmed observations own a retracement range.
Untouched definition	The exact event that changes an untouched level to touched must be objective and timestamped.
Round-level significance	Lookback, tolerance, minimum interactions and minimum reaction size are undefined.
Session-end behavior	The lifecycle of unfinished states at the session boundary remains unspecified.
Within-candle ambiguity	OHLC does not reveal the internal order of multiple price interactions.
Costs	Spread, slippage and commission assumptions are absent.
Volume imbalance	The dataset may not contain centralized traded volume; price imbalance and volume imbalance are not interchangeable.

5. Conceptual Knowledge Map

These definitions are academic and non-operational. They explain what the terms mean without specifying trade actions or realistic execution.

Fair Value Gap

A label for a three-observation price discontinuity in which the outer observations' ranges do not overlap. The label depends on precisely defined indexing, timestamps and zone boundaries.

BISI

The bullish orientation of a three-observation range discontinuity. A research implementation must distinguish formation time from the later time at which the pattern becomes knowable.

SIBI

The bearish inverse of BISI. The same requirements for objective boundaries, formation, interaction, expiration and chronological access apply.

Imbalance lifecycle

A zone may be modeled as formed, untouched, interacted with, traversed, invalidated or expired. Every transition requires an exact data condition and timestamp.

ATR

A rolling volatility statistic that incorporates a period's high-low movement and its relationship to the preceding close. It measures movement magnitude, not direction. Lookback length, warm-up policy and missing-period behavior must be declared.

Volume imbalance

A statement about asymmetric traded activity. It requires a trustworthy volume field and an objective definition. Forex OHLC alone does not prove true traded-volume imbalance.

Look-ahead bias

Using information before it was historically knowable. A centered pattern that needs later observations cannot be emitted at its center timestamp.

Event time vs confirmation time

Event time records where a feature appears in the series; confirmation time records when all necessary evidence has arrived. Sequential research must consume only the latter.

6. Safe Detailed Engineering Roadmap

The next work should harden reproducibility, portability, integration testing, and generic chronological processing before any age-restricted strategy simulation is considered.

Phase A - Prevent accidental overwrite

- Separate sample and production paths.
- Add a test proving the paths differ.
- Require an explicit overwrite flag for existing production files.
- Keep samples, raw data and processed data outside Git.

Phase B - Generalize instrument cleaning

- Move precision, pip/tick size and repair tolerance into instrument metadata.
- Pass metadata into the cleaner rather than importing an EUR/USD constant.
- Reject an unsupported symbol explicitly.
- Create boundary tests per supported metadata profile.

Phase C - Dataset manifests

- Record symbol, interval, source, row count, start, end, gap count, repair count and rejection count.
- Calculate a SHA-256 file hash.
- Record the cleaning-policy version and creation time.
- Write the manifest beside the processed dataset.

Phase D - Chunk integrity

- Compare chunk filenames with their internal timestamp ranges.
- Detect empty, corrupt, overlapping or unsorted chunks.
- Compare duplicate rows before choosing which one to keep.
- Produce a machine-readable integrity report.

Phase E - Retry and failure classification

- Separate timeout, rate limit, authentication, invalid request and earliest-boundary outcomes.
- Use bounded retries with increasing delay only for transient failures.
- Never retry authentication or malformed requests indefinitely.
- Preserve every completed chunk before retrying.

Phase F - Full-dataset integration tests

- Validate schema, UTC dtype, order, uniqueness and cleaned OHLC invariants.
- Check an expected date and row-count range.
- Verify every accepted 15-minute row has 15 source rows.
- Run integration tests separately from fast unit tests.

7. Generic Chronological Framework

A reusable, non-trading time-series engine can be developed and tested with synthetic data. Its purpose is to prove chronological access and auditable state transitions.

Component	Responsibility
TimestampedObservation	Immutable data item with timestamp and payload
Event	Typed statement emitted from information currently available
StateTransition	Old state, new state, timestamp and reason
EventLog	Append-only record used for reconciliation
SequentialEngine	Rejects time reversal and processes one observation at a time

Acceptance requirements

- No component can request rows after the current engine timestamp.
- Out-of-order input raises an explicit error.
- Every state change is logged with a deterministic reason.
- Synthetic tests prove delayed confirmation behavior.
- Repeated input produces identical output.

8. Generic Statistics and Visualization

Safe research features include candle range, absolute close-to-close change, rolling mean, rolling standard deviation, data coverage, gap distribution and missing-period heatmaps. Each rolling statistic should declare its input, lookback, minimum observations, warm-up behavior and treatment of missing periods.

A read-only viewer can display one-minute and 15-minute candles, UTC and New York times, missing-data markers, repaired/rejected-row markers, and the constituent source rows of a selected 15-minute period. It should not display buy/sell prompts or simulated orders.

9. Recommended Next Six Stories

Order	Story	Estimate	Acceptance checkpoint
1	Separate sample and production filenames	1 h	Sample test cannot overwrite full history
2	Remove obsolete backups	0.5 h	Git history becomes the only version archive
3	Move tolerance into instrument metadata	2 h	Cleaner has no EUR/USD-specific global
4	Create manifest and SHA-256 hash	2 h	Every processed file has reproducible metadata
5	Build chunk-integrity validator	2 h	Corrupt and overlapping chunks are reported
6	Add real-dataset integration suite	2 h	Data invariants pass outside the unit suite

10. Acceptance Gates

Gate	Current state	Requirement
Data gate	Provisional pass	Timezone and resampling are verified; portability and manifests remain
Swing gate	Not started	Manual samples and delayed confirmation must be proven
Structure gate	Not started	All state transitions must be deterministic
Pattern gate	Not started	Formation and lifecycle order must be validated
Volatility gate	Not started	Definition, period and missing-data behavior must be fixed
Simulation gate	Outside this report	Requires qualified adult or instructor supervision
Trust gate	Not started	Selected history must be reconciled event-by-event

11. Risks and Controls

Risk	Control
Secret leakage	Keep keys in .env, sanitize request errors, rotate exposed credentials
Sample overwrite	Separate paths and require explicit overwrite authorization
Look-ahead	Confirmation timestamps, sequential processing and future-access tests
Incorrect alignment	Constituent-row reconciliation and boundary tests
Hidden assumptions	Versioned assumptions register and configuration validation
Instrument mismatch	Per-instrument metadata and explicit unsupported-symbol errors
Data drift	Manifests, hashes and versioned cleaning policy
Provider gaps	Report and preserve missing periods; never fabricate candles

12. Final Assessment

The project has achieved a strong data-engineering foundation beyond the original Sprint 1 data scope. It can download, resume, merge, clean, diagnose, resample and test a large EUR/USD history. The decisive next milestone in the original blueprint would be chronological feature confirmation, but the immediate engineering priority should be reproducibility, generalization, integrity checks and integration testing.

Current overall status: Sprint 1 approximately 75%; complete blueprint approximately 15-20%; data foundation approximately 85-90%; strategy and execution functionality not implemented.

Scope note: this document supports historical data engineering and educational analysis only. It does not provide realistic trade-entry, stop, target, execution, P&L; or live brokerage instructions.